

Spring 2008

Problem 1 (Sp08). Prove that it is not possible to find two linear operators A and B on a nonzero finite dimensional complex vector space with $AB - BA = I$, where I is the identity operator. Give an example of two such operators acting on an infinite dimensional complex vector space.

Problem 2 (Sp08). Evaluate $\int_{-\infty}^{+\infty} \frac{\cos x}{1+x^2} dx$.

Problem 3 (Sp08). Classify the subgroups of the symmetric group S_4 .

Problem 4 (Sp08). Find the solution of the differential equation

$$y'' - 2y' + y = e^{-x}$$

satisfying $y(0) = y'(0) = 0$.

Problem 5 (Sp08). Suppose M is an $n \times n$ nilpotent matrix over $\mathbb{C}$. Show the set of matrices $C(M)$ which commute with M is the ring $\mathbb{C}[M]$ if and only if the null space of M has dimension one.

Problem 6 (Sp08). Suppose G is a finite group with only one automorphism. Show that $|G| \leq 2$.

Problem 7 (Sp08). Find all irreducible polynomials of degree at most 4 over the field with 2 elements.

Problem 8 (Sp08). Let p, q be distinct prime numbers and let R be a commutative ring with identity of characteristic pq . Show there are rings S, T of characteristic p, q respectively, such that R is isomorphic to $S \times T$.

Problem 9 (Sp08). For integers $n \geq 1$, let $f(n)$ stand for the maximum order of elements of S_n . Show that

$$\liminf_{n \rightarrow \infty} \frac{n}{f(n)} = 0$$

Problem 10 (Sp08). For integers $n \geq 1$, let P_n denote the set of polynomials of degree at most n and real coefficients. Show there is $q(x) \in P_n$ such that for all $p(x) \in P_n$

$$\int_0^1 p(x)q(x) dx = \int_0^1 \frac{p(x)}{x^2 + 1} dx$$

Problem 11 (Sp08). 1. Let G be a finite abelian group, and let c be the product of all elements of G . Show that $c^2 = 1$.

2. Let $\mathbb{F}$ be a finite field, and let c be the product of all nonzero elements in $\mathbb{F}$. Show that $c = -1$.

Problem 12 (Sp94, Sp08). Let G be a group and $H \subset G$ a subgroup of finite index n . Show that G contains a normal subgroup N such that $N \subset H$ and the index of N is at most $n!$.

Problem 13 (Sp08). Let p and q be distinct primes. Show that any group G of order p^2q^2 is not simple.

Problem 14 (Sp08). Let $\zeta = e^{2\pi i/5}$ and let $\alpha = \sqrt[5]{2} \in \mathbb{R}$. Let $\mathbb{E}$ denote the subfield $\mathbb{Q}[\zeta, \alpha] \subset \mathbb{C}$ generated by ζ and α . Show that $\mathbb{E}$ is Galois over $\mathbb{Q}$. What is $[\mathbb{E} : \mathbb{Q}]$?

Problem 15 (Sp08). $y(x)$ defined on $[0, \infty)$ satisfies $y'' - y = f(x)$ in $x > 0$, $y(0) = 0$, $y'(0) = 0$. Here $f(x)$ is a continuous function on $[0, \infty)$ which vanishes for $x > 1$. If $y(x)$ and $y'(x)$ tend to 0 as $x \rightarrow \infty$ show that $\int_0^1 f(x)e^{-x} = 0$.

Problem 16 (Sp08). Evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^5}$$

Problem 17 (Sp08). Compute the limit of the sequence (x_n) of real numbers so that

$$x_n = x_{n-1} - \frac{1}{2}x_{n-2}$$

for $n \geq 2$, and $x_0 = x_1 = 1$.

Problem 18 (Fa06, Sp08). Compute $\lim_{x \rightarrow 0} \frac{d^4}{dx^4} \frac{x}{\sin x}$.